Chess v1 — Cut Plan

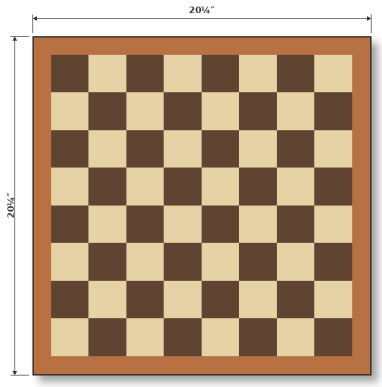
One end-grain chess-pattern board, walnut and brown maple 2.25 in cells, cherry border 1.125 in all sides, finished 20.25 in square, 0.5 in thick. Each row is built as two half-width panels split at the centre line (left half mirrors right half), so no panel is wider than 10.125 in and every panel fits the 12.5 in planer. Only three panel patterns are built, A, B and C, and both A and B start with a cherry component. Their mirrors are the same slices placed flipped (flip H), A flipped covers the old BR half, B flipped covers the old AR half, C flipped covers the old CR half.

Cut list for the finished design, in build order.

1. Warnings

#	Lvl	Type	Message
1	WARNING	Small Margin	Board BW1A leaves only ⅛″ margin (< ¼″) — too tight; prone to under-supplying strips if the board runs narrow.
2	WARNING	Zero Offcut	6 strip(s) crosscut to an exact fit with 0″ offcut — no margin for cutting error in the crosscut phase: BM1A ×2, BC1A ×1, BC2A ×1, BW1A ×2. A cut even slightly long, or kerf drift, will make the last piece short. Consider longer blanks or fewer pieces per strip.

2. Finished board pattern



3. Volume Sanity Check

Review the per-species volumes against inventory.

Species	Available	Required	Volume Offset
maple	4.7 BF	0.6 BF + 1.5 BF loss = 2.0 BF	+2.7 BF (235%)
cherry	2.6 BF	0.3 BF + 0.8 BF loss = 1.1 BF	+1.5 BF (231%)
walnut	2.7 BF	0.6 BF + 1.5 BF loss = 2.0 BF	+0.7 BF (134%)

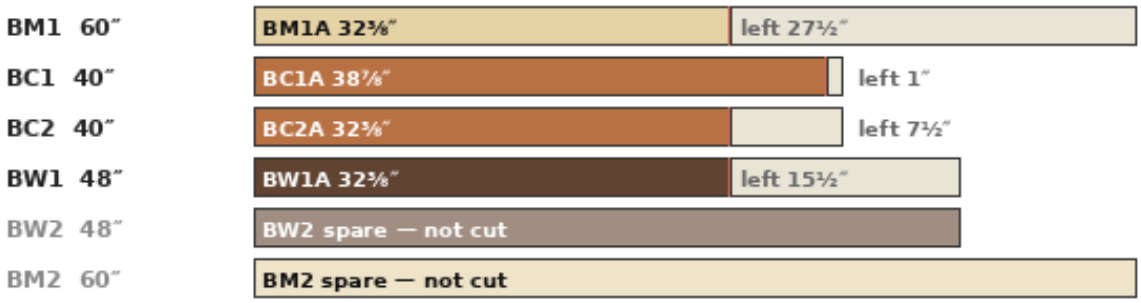
4. Crosscut Blanks

This plan involves cutting blanks from the inventory — review the inventory and the planned cuts.

Raw board	Species	Raw length	Blanks (id = length)	Leftover
BM1	maple	60″	BM1A = 32⅜″	27½″

Raw board	Species	Raw length	Blanks (id = length)	Leftover
BC1	cherry	40″	BC1A = 38⅞″	1″
BC2	cherry	40″	BC2A = 32⅜″	7½″
BW1	walnut	48″	BW1A = 32⅜″	15½″
BW2	walnut	48″	— (untouched spare)	—
BM2	maple	60″	— (untouched spare)	—

Crosscut Diagram



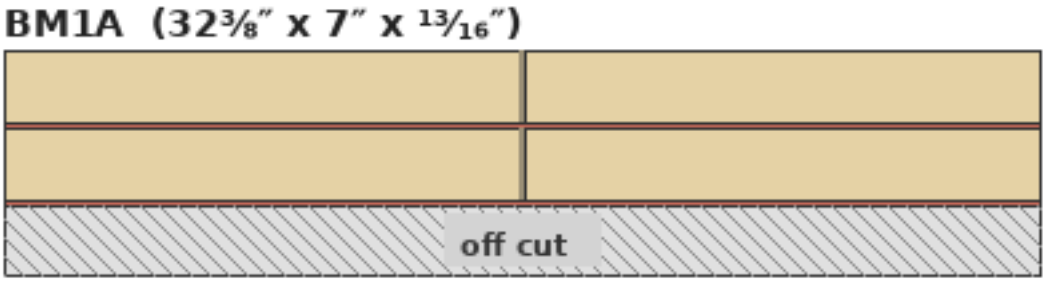
5. Initial board rips (the cut list)

Review the rip list.

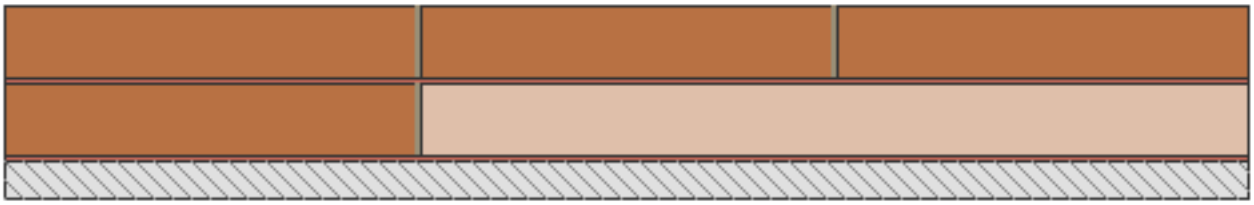
Board	Origin	Species	Length	Rips (width × count)	Used	Loss	Margin
BM1A @ 7″	BM1	maple	32⅜″	2¼″ ×2	4½″	⅜″	2⅛″
BC1A @ 6″	BC1	cherry	38⅞″	2¼″ ×2	4½″	⅜″	1⅛″
BC2A @ 5½″	BC2	cherry	32⅜″	1⅛″ ×2	2¼″	⅜″	2⅞″
BW1A @ 5″	BW1	walnut	32⅜″	2¼″ ×2	4½″	⅜″	⅛″

Board Rip Diagram

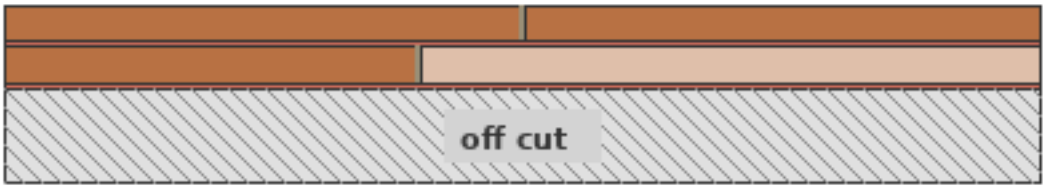
Review the board rip diagrams against the rip list.



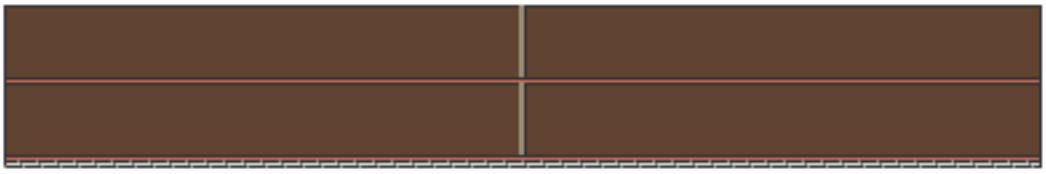
BC1A (38⅞″ x 6″ x 13⁄16″)



BC2A (32⅜″ x 5½″ x 13⁄16″)



BW1A (32⅜″ x 5″ x 13⁄16″)

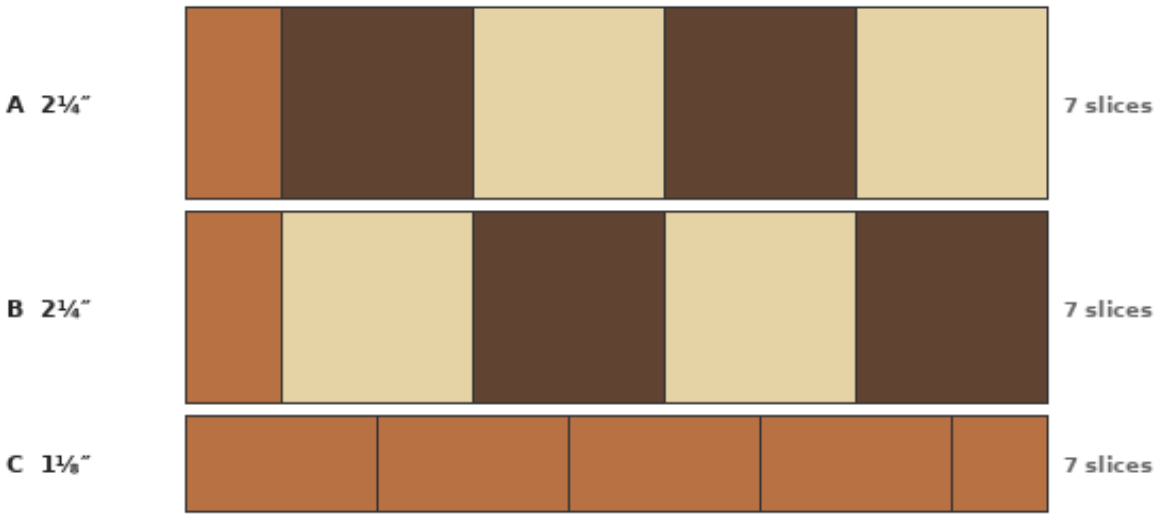


6. Panels overview

Panel	Qty	Length	Slices	Height	Ripped	Sequence (outer → in)
A	1	16⅛″	9	2¼″	—	CB · W · M · W · M
B	1	16⅛″	9	2¼″	—	CB · M · W · M · W
C	1	12⅞″	7	1⅛″	—	CT · CT · CT · CT · CTH

Panel End Grain Diagram

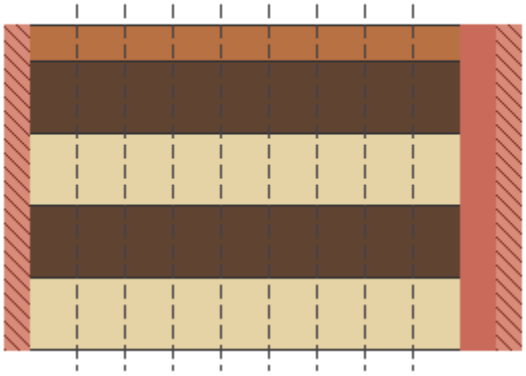
Review the panel end-grain diagrams.



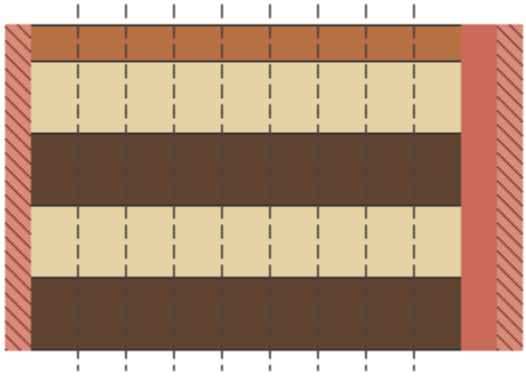
Panel Face Diagram

Review the panel face diagrams.

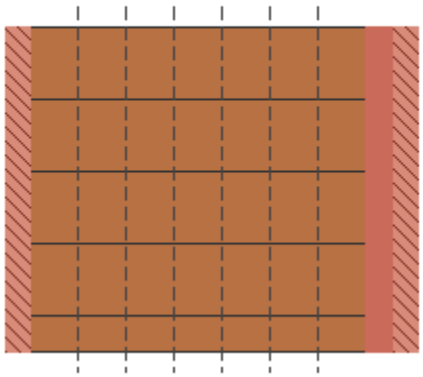
Panel A ×1
16⅛″ long (9 slices)

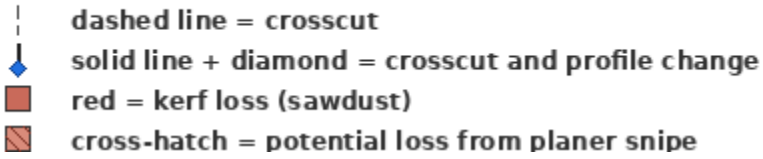


Panel B ×1
16⅛″ long (9 slices)



Panel C ×1
12⅞″ long (7 slices)





Slice Fulfillment

Review slice demand, supply, and offset.

Slice	Demand	Supplied	Supply Offset
C	4 × 1 = 4	7	+3
A	8 × 1 = 8	9	+1
B	8 × 1 = 8	9	+1

7. Components

Review component demand, fulfillment, and shortfall.

Component	Species	Width	Length	Demand	Fulfilled	Shortfall
CB	cherry	1⅛″	16⅛″	2	2	0
CT	cherry	2¼″	12⅞″	4	4	0
CTH	cherry	1⅛″	12⅞″	1	1	0
M	maple	2¼″	16⅛″	4	4	0
W	walnut	2¼″	16⅛″	4	4	0

8. Panels (profiles and sequence)

Review the panels and their quantities.

Panel	Qty	Length	Profiles (slices)	Component sequence
A	1	16⅛″	single: 9	CB · W · M · W · M
B	1	16⅛″	single: 9	CB · M · W · M · W
C	1	12⅞″	single: 7	CT · CT · CT · CT · CTH